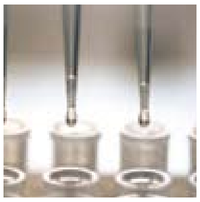
SELDI Technology



ProteinChip® SELDI Standardization Suite

Confidently Achieve Consistent, Reproducible Results

BIO-RAD

ProteinChip SELDI Standardization Suite

Confidently Achieve Consistent, Reproducible Results

Laboratories involved in protein biomarker research that use cutting-edge technologies, such as mass spectrometry, are often confronted with reproducibility issues related to system variability, operation complexity, lack of training, and unfriendly software interface.

The ProteinChip SELDI standardization suite is a family of products designed to empower ProteinChip SELDI users and ensure consistent, reproducible results over time, operators, and locations.

The ProteinChip SELDI standardization suite empowers ProteinChip SELDI system users to:

- Train and acquire reproducible data with confidence
- Validate the system and its performance
- Upgrade the instrument

ProteinChip SELDI System Components Upgrade

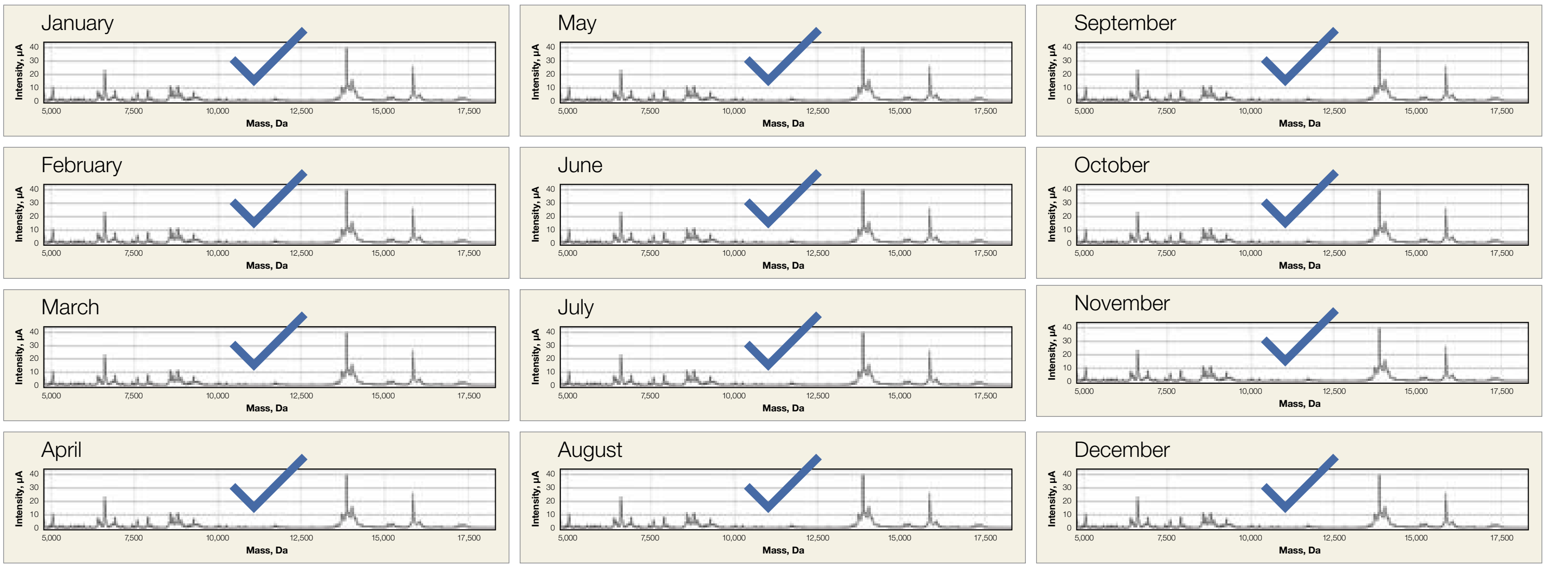
ProteinChip OQ Kit and System Check Kit

ProteinChip Peptide Mass Calibration Kit

ProteinChip Data Manager Software 3.5

ProteinChip Detector Calibration Kit

ProteinChip System Starter Kit



The ProteinChip SELDI standardization suite ensures consistent, reproducible results.

The ProteinChip SELDI Process

The ProteinChip SELDI system uses surface-enhanced laser desorption/ionization (SELDI) technology to detect and calculate the mass of proteins and peptides. The system integrates ProteinChip arrays, a ProteinChip SELDI reader, and ProteinChip SELDI system software to analyze samples based on measured time-of-flight. Compact enough to fit into almost any laboratory, the ProteinChip SELDI system enables direct access to precise mass analysis of peptides and proteins from complex biological samples.

The ProteinChip SELDI system is an ideal tool to compare samples in fast, high-throughput experiments for differential expression protein analysis. Its most common application is to profile complex samples, which is the first step in protein biomarker discovery.

Key Features of the Standardization Suite

- **Comprehensive package to improve reproducibility** — the ProteinChip SELDI standardization suite empowers SELDI system users to update their instrument, validate system performance, train using a step-by-step approach, and, ultimately, improve the quality of their results
- **Upgrade and replacement of critical components and subassemblies** — the ProteinChip SELDI system components upgrade optimizes instrument performance and improves system consistency and reliability
- **Complete system evaluation and optimization** — ProteinChip qualification and calibration kits evaluate system performance and ensure that the instrument is operating at specifications
- **Updated software** — ProteinChip data manager software 3.5 simplifies system evaluation procedures, reduces the risk of errors, and provides greater confidence in data generated by the ProteinChip SELDI system
- **Training using a step-by-step approach** — the ProteinChip system starter kit includes a complete set of ready-to-use consumables, and is a powerful training tool for new customers and for SELDI users who wish to improve their experimental technique

Train

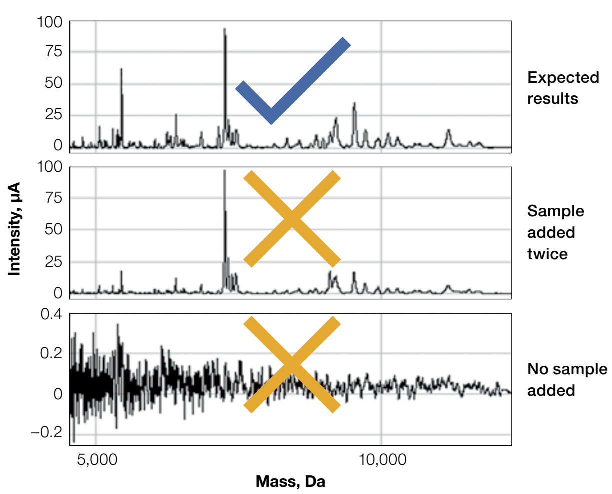
> **ProteinChip System Starter Kit**

The ProteinChip system starter kit has been redesigned to deliver a step-by-step focused introduction to ProteinChip SELDI technology. The kit is an effective training tool for customers, and can also be used in conjunction with the ProteinChip SELDI system basic course. The kit minimizes external variables that are not associated with the user (for example, sample, lot, and reagent variation), in order to focus on evaluating and improving experimental techniques prior to starting a research project. The starter kit includes an in-depth video tutorial, and the ProteinChip arrays, buffer, and reagents needed to generate and evaluate protein profiles from the included human serum and *E. coli* samples. After customers become familiar with instrument operation and data acquisition, they can use the analysis tools provided in the kit to:

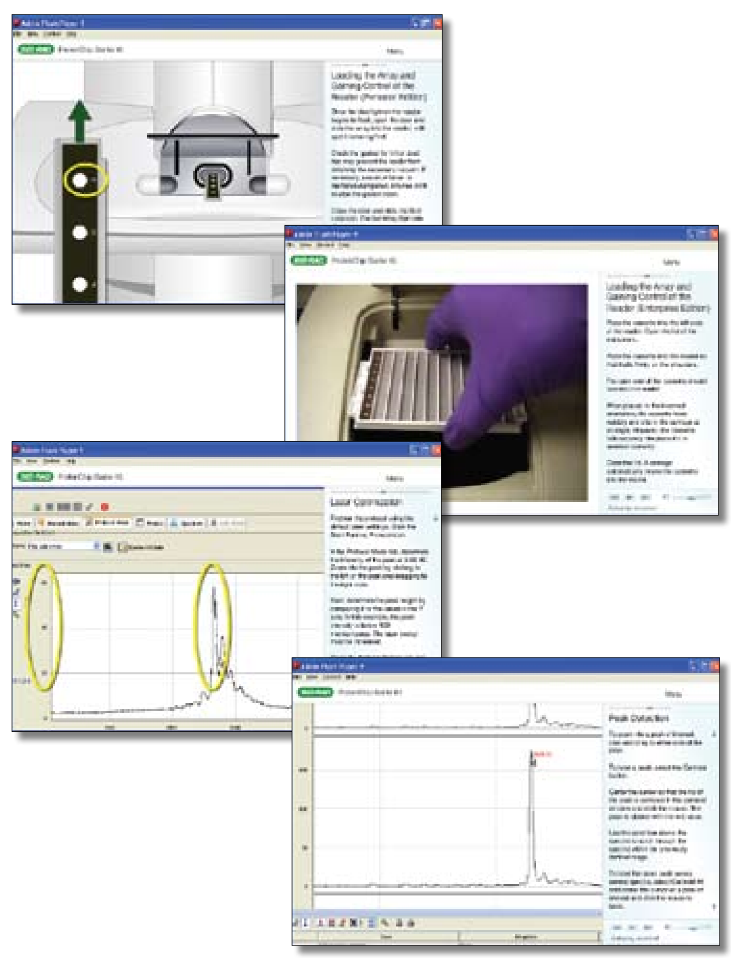
- Determine the reproducibility and optimal peak count of protein profiles
- Compare results to the included premade ProteinChip NP20 and CM10 arrays (premade ProteinChip arrays are prepared using the same samples and reagents included in the kit)
- Improve results by employing the valuable tips and techniques outlined in the manual

Learn the Basics Fast With a Video Tutorial

The video tutorial included with the starter kit illustrates the basic technology of the ProteinChip SELDI system. It starts with a visual description of the anatomy of a ProteinChip array and ends with instructions for performing mass calibrations. Watch the video to become familiar with the concepts, then perform the same tasks using the ProteinChip peptide standard array and instructions included with the kit. The video is also available online at **www.bio-rad.com/proteinchip/starterkit/**.



Spectra examples, as included in the manual, help to identify possible technical errors that can cause problems. Data illustrate the results of an error made in the number of additions of sample to the CM10 array during the binding step.



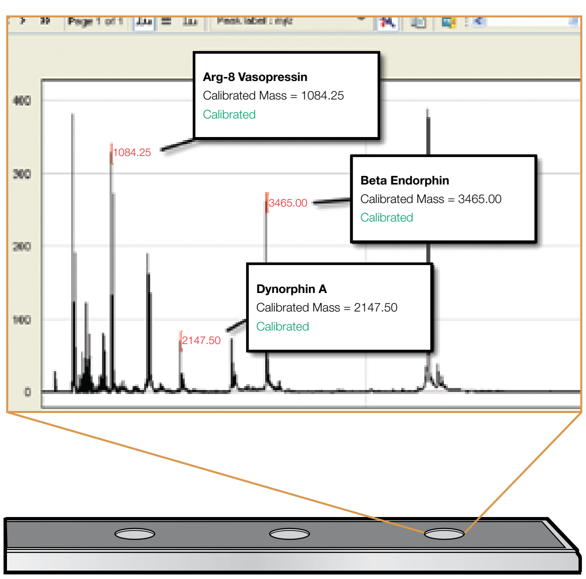
Validate

ProteinChip OQ Kit and ProteinChip System Check Kit

The ProteinChip OQ and system check kits are designed for use in regulated laboratories. Use these kits to generate data on system performance and to create reports that show reader compliance to operational specifications. No sample preparation is required; ready-to-use ProteinChip detector calibration arrays are included in the kits.

ProteinChip Peptide Mass Calibration Kit

The ProteinChip peptide mass calibration kit contains a ProteinChip peptide standard array that has been prepared with matrix and a mix of seven different peptides ranging in molecular weight from 1,084 to 12,230. The kit allows up to 160 external mass calibrations within a 12-week period in the peptide region; a typical calibration takes as little as 10 minutes. Because the array is ready-to-use with no preparation steps required, the kit is easy to use and saves time.



Peak	Name	Molecular Weight
1	Arg-8 vasopressin	1084.247
2	Somatostatin	1637.903
3	Dynorphin A	2147.5
4	ACTH	2933.5
5	Beta endorphin	3465
6	Arg-insulin	5963.8
7	Cytochrome C	12230.92

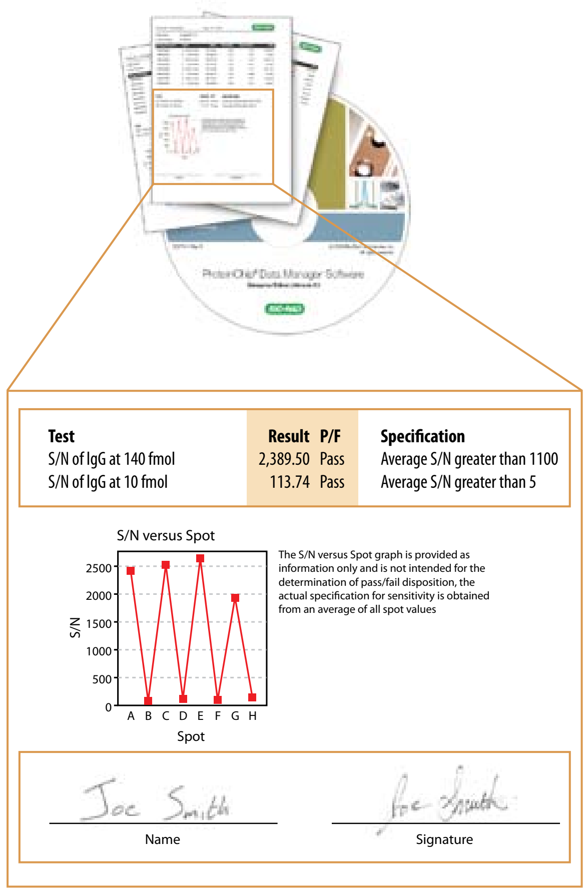
The ProteinChip peptide mass calibration kit provides a surface that contains seven peptides and ready-to-use matrix.

ProteinChip Detector Calibration Kit

Over time, some components common to all mass spectrometers may alter system performance. For example, detector output changes over time, producing variability in signal-to-noise ratios and peak intensity values. This kit includes a ProteinChip detector calibration array, which provides data that the ProteinChip SELDI reader uses to automatically adjust the detector voltage. These adjustments stabilize the detector's output, maintaining data reproducibility.

ProteinChip Data Manager Software 3.5

Version 3.5 simplifies the calculation of OQ results by automatically applying the correct processing parameters, labeling the required peaks, and generating a PDF report of the results, all from a single mouse click. The new software also automatically generates qualification and calibration reports.



ProteinChip data manager software 3.5 includes a menu option for creating reports, which can be applied to data generated from the ProteinChip OQ and system check kits.

Upgrade

ProteinChip Data Manager Software 3.5

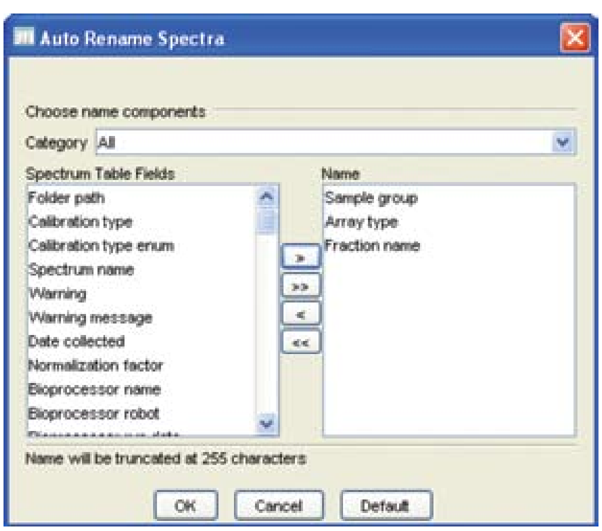
Version 3.5 is the latest release of the ProteinChip data manager software, the ProteinChip SELDI system's integrated instrument control and data analysis software solution. The data manager software controls data acquisition from the ProteinChip SELDI reader and includes a robust client-server architecture coupled with a relational database system for managing and tracking SELDI data. With ProteinChip data manager software 3.5 you can:

Automatically generate qualification and calibration reports
Version 3.5 simplifies the calculation of OQ results by automatically applying the correct processing parameters, labeling the required peaks, and generating a PDF report of the results, all from a single mouse click.

Save spectrum analysis parameters
With version 3.5, SELDI users can save a suite of analysis parameters to a file, and recall and apply them to another data set. This time-saving feature can reduce repetition and potential errors.

Dynamically name spectra
The new spectrum naming feature enables users to change a spectrum name in an experiment from the default name, to one with user-specified sample properties.

Rapidly set up high-voltage reconditioning protocols
Automatic protocol setup for high-voltage reconditioning streamlines maintenance for the ProteinChip SELDI system.



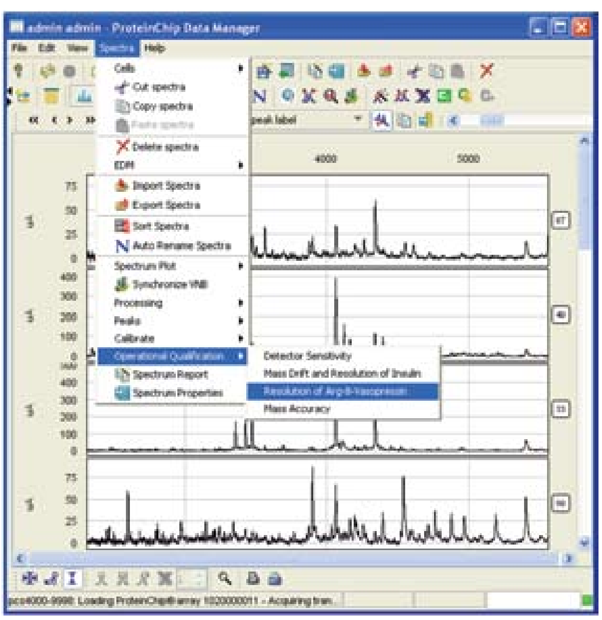
Dynamic spectrum naming. Automatically change a spectrum name from the default name, to one with user-specified sample properties.

ProteinChip SELDI System Components Upgrade

Optimize your instrument performance and improve productivity with the ProteinChip SELDI system components upgrade. Bio-Rad offers an advanced system update and preventive maintenance service for your ProteinChip SELDI system that include:

- Complete system evaluation and diagnostics
- System calibration and optimization
- Servicing of all moving parts
- Optical path alignments
- Signal path electronic adjustments
- Replacement of critical components and subassemblies if found defective or at a lower revision than current factory specifications

Ensuring that your ProteinChip SELDI system is performing at specifications is the first step to achieving consistent, reproducible results.



One-click Operational Qualification report generation. ProteinChip data manager software 3.5 ensures consistent processing of qualification data.

Ordering Information

Catalog #	Description
Standardization Suite Components	
C70-00068	**ProteinChip System Starter Kit**, includes 3 each of CM10 and NP20 ProteinChip arrays, 1 each of CM10 and NP20 ProteinChip premade arrays, 1 ProteinChip peptide standard array, 2 vials of ProteinChip SPA matrix, ProteinChip CM low-stringency buffer, acetonitrile, human serum, *E. coli*, 1% TFA, CD-ROM with video tutorial and protocols, instructions
C70-00070*	**ProteinChip Peptide Mass Calibration Kit**, includes 1 ProteinChip peptide standard array, sufficient for up to 160 calibrations, CD-ROM with video tutorial and protocol, instructions
C70-00080	**ProteinChip OQ Kit**, includes 2 detector calibration arrays, 6 detector qualification arrays, 2 peptide standard arrays, CD with protocols, instructions
C70-00081	**ProteinChip System Check Kit**, includes 1 detector calibration array, 1 detector qualification array, 1 peptide standard array, CD with protocols, instructions
C70-00082	**ProteinChip Detector Calibration Kit**, includes 1 detector calibration array, instructions
SW3-020050	**ProteinChip Data Manager Software 3.5, Add 1 User to Network License**
SW3-040010	**ProteinChip Data Manager Software 3.5, Personal Edition**, includes 1-user license
SW3-040030	**ProteinChip Data Manager Software 3.5, Enterprise Edition**, includes 5-user network license
Z50-00015APM	**ProteinChip SELDI System, Components Upgrade, Enterprise Edition**, system evaluation and components upgrade
Z50-00016APM	**ProteinChip SELDI System, Components Upgrade, Personal Edition**, system evaluation and components upgrade

Catalog #	Description
Systems	
ZO5-10014	**ProteinChip SELDI System Upgrade**, upgrades Personal Edition to Enterprise Edition
Z50-00011	**ProteinChip SELDI System, Enterprise Edition**, 115 V
Z50-00013	**ProteinChip SELDI System, Personal Edition**, 115 V
Z50-00015	**ProteinChip SELDI System, Enterprise Edition**, 115 V, with computer and monitor
Z50-00016	**ProteinChip SELDI System, Personal Edition**, 115 V, with computer and monitor
Z50-00021	**ProteinChip SELDI System, Enterprise Edition**, 230 V
Z50-00023	**ProteinChip SELDI System, Personal Edition**, 230 V
Z50-00025	**ProteinChip SELDI System, Enterprise Edition**, 230 V, with computer and monitor
Z50-00026	**ProteinChip SELDI System, Personal Edition**, 230 V, with computer and monitor
Accessories	
A10-20012	**Computer and Monitor**, for ProteinChip SELDI systems
A30-00010	**Bar Code Scanner**, handheld
A30-10010	**Uninterruptible Power Supply**, 120 V, for Protein Chip SELDI systems
A30-10020	**Uninterruptible Power Supply**, 230 V, for Protein Chip SELDI systems

* The ProteinChip peptide standard array has a 12-week shelf life after opening.

The SELDI process is covered by U.S. patents 5,719,060; 6,225,047; 6,579,719; and 6,818,411 and other issued patents and pending applications in the U.S. and other jurisdictions.

Bio-Rad Laboratories, Inc.

Life Science Group

Web site *www.bio-rad.com* ***USA*** *800 4BIORAD* ***Australia*** *61 02 9914 2800* ***Austria*** *01 877 89 01* ***Belgium*** *09 385 55 11* ***Brazil*** *55 21 3237 9400* ***Canada*** *905 364 3435* ***China*** *86 21 6426 0808* ***Czech Republic*** *420 241 430 532* ***Denmark*** *44 52 10 00* ***Finland*** *09 804 22 00* ***France*** *01 47 95 69 65* ***Germany*** *089 318 84 0* ***Greece*** *30 210 777 4396* ***Hong Kong*** *852 2789 3300* ***Hungary*** *36 1 455 8800* ***India*** *91 124 4029300* ***Israel*** *03 963 6050* ***Italy*** *39 02 216091* ***Japan*** *03 6361 7000* ***Korea*** *82 2 3473 4460* ***Mexico*** *52 555 488 7670* ***The Netherlands*** *0318 540666* ***New Zealand*** *0508 805 500* ***Norway*** *23 38 41 30* ***Poland*** *48 22 331 99 99* ***Portugal*** *351 21 472 7700* ***Russia*** *7 495 721 14 04* ***Singapore*** *65 6415 3188* ***South Africa*** *27 861 246 723* ***Spain*** *34 91 590 5200* ***Sweden*** *08 555 12700* ***Switzerland*** *061 717 95 55* ***Taiwan*** *886 2 2578 7189* ***United Kingdom*** *020 8328 2000*

Bulletin 5799 Rev A US/EG *08-0838 1208 Sig 0308*